Bioinformatic analysis of the active site diversity for SARS-CoV-2 and other coronaviruses

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OVERVIEW

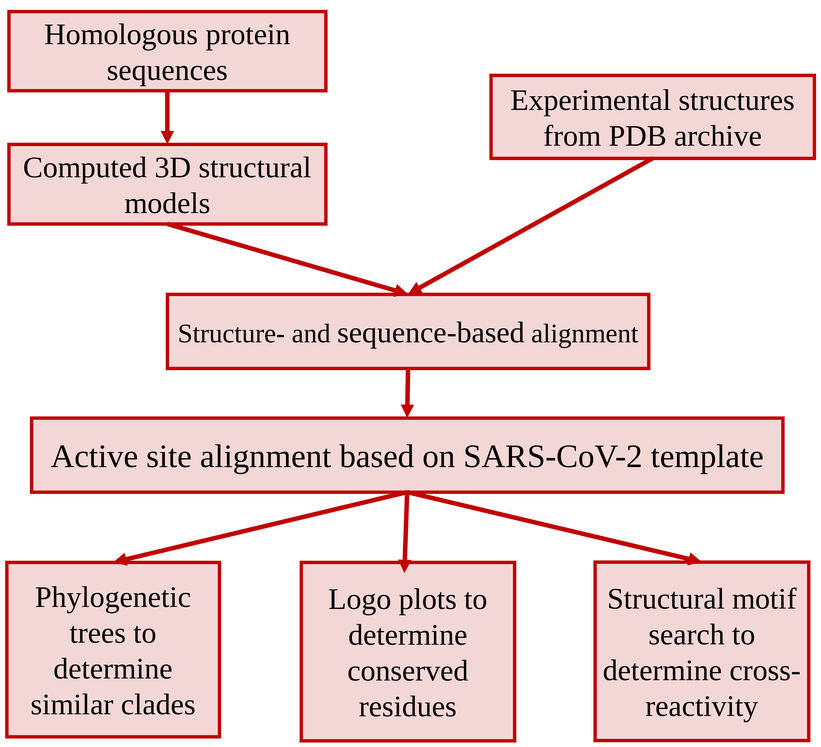
Introduction:

- Coronaviruses are a significant risk to global health
- Over four million lives have been lost to SARS-CoV-2 (COVID-19)
- Treatments available for SARS-CoV-2 and other coronavirus infections in people is limited
- Proteases are essential to the processing of viral proteins

Objectives:

- Identify conserved enzyme active sites in proteases across coronaviruses to aid in broad-spectrum drug development

METHODOLOGY



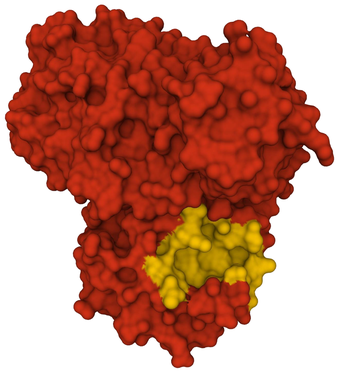
DESCRIPTION OF PROGRAMS

Mol*: Web-based visualization tool for 3D modeling and manipulation of molecular data. https://molstar.org/
Clustal Omega: Web-based multiple sequence alignment tool. https://www.ebi.ac.uk/Tools/msa/clustalo/
Dali: 3D protein structure comparison server. http://ekhidna2.biocenter.helsinki.fi/dali/
Robetta: Protein structure prediction service. https://robetta.bakerlab.org/
WebLogo: Logo plot generator. https://weblogo.berkeley.edu/logo.cgi
MEGA-X: Software to generate phylogenetic trees. https://www.megasoftware.net/

RESULTS

Main Protease (MPro)

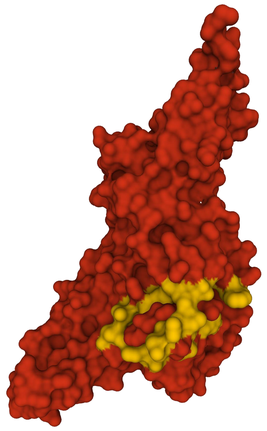
- 368 homologous proteins analyzed
- 27 residues identified as part of the active site
- 11 residues completely conserved in all MPro homologs
- No human proteins with similar active sites in the Protein Data Bank



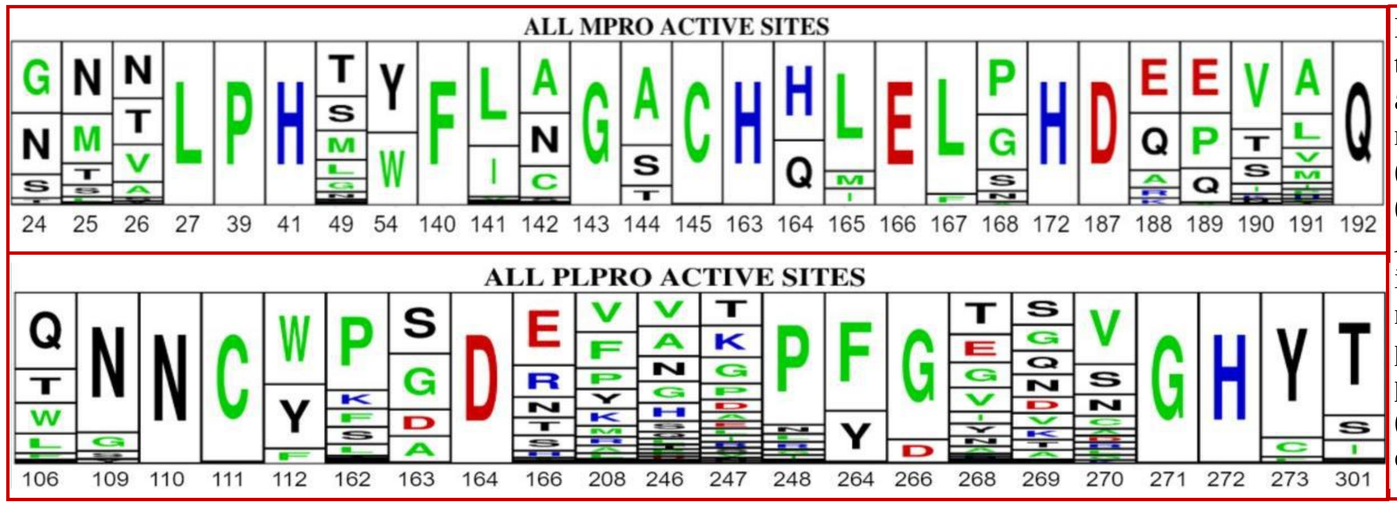
3D Model of SARS-CoV-2 MPro (PDB ID: 6YB7). Active site is in yellow.

Papain-Like Protease (PLPro)

- 268 homologous proteins analyzed
- 22 residues identified as part of the active site
- 5 residues completely conserved in all PLPro homologs
- No human proteins with similar active sites in the Protein Data Bank



3D Model of SARS-CoV-2 PLPro (PDB ID: 6WUU). Active site is in yellow.



Logo plots displaying the frequency of amino acids at each active site residue in all MPro (*top*) and PLPro (*bottom*) homologs. Amino acid groups are indicated by colors: non-polar (green), polar (black), positively charged (blue), negatively charged (red).

CONCLUSIONS

The highly conserved residues present in the active sites of Coronaviridae main and papain-like proteases indicate that a broad-spectrum drug can be used to target these essential enzymes. In the future, a small molecule drug can inhibit these proteases and prevent viral protein processing ultimately leading to limited viral spread and recovery from infection. The drug screening process is underway at the Department of Molecular Design and Synthesis at Rutgers University.

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RUTGERS

RCSB PDB PROTEIN DATA BANK

IQB

UMBC